PTL AI — Play Teach Learn

Imercfy Private Limited

Product Requirements Document

“Tutor Me” — Interactive AI Tutoring Session

Triggered from Lesson Shorts Video Player

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1. Executive Summary

This document defines the requirements for the “Tutor Me” feature — a new interactive AI tutoring mode triggered from within PTL AI’s Lesson Shorts video player. When a student selects “Tutor Me,” the video pauses and an AI-driven tutoring session begins, powered by Miss Lily (the PTL AI personal tutor avatar). The tutor walks the student through each video segment one at a time, using the exact same images, animations, and text content stored in the Smart Content Studio database that was used to generate the original video.

After explaining each segment, the tutor prompts the student to repeat and demonstrate their understanding. If the student struggles, the tutor adapts its teaching approach — using simpler language, analogies, or alternative visual aids from the content database — until the student can replicate the concept without errors. This segment-by-segment cycle continues through the entire lesson, culminating in a final comprehensive assessment where the student must articulate the full lesson in their own words.

This feature directly addresses the core problem PTL AI solves: passive video consumption leads to zero retention. “Tutor Me” transforms every Lesson Short from a one-way broadcast into an active, personalised retrieval practice session — the single most effective learning method backed by four decades of cognitive science research.

2. Objectives & Success Metrics

2.1 Business Objectives

- Increase student concept retention by 40–60% compared to passive video-only consumption
- Deepen engagement time per lesson by introducing active interaction loops within existing content
- Strengthen the value proposition of Miss Lily AI Tutor across all subscription tiers
- Generate measurable mastery data per student per concept for the Teacher Analytics Dashboard
- Create a competitive differentiator that no other Indian EdTech platform offers

2.2 Success Metrics

Metric	Target	Measurement Method
Segment-level comprehension pass rate	≥75% on first attempt	AI evaluation score per segment
Full-lesson comprehension pass rate	≥60% on first attempt	Final assessment AI evaluation
Student re-engagement rate	+30% return within 48hrs	Session analytics
Average session completion rate	≥70%	Tutor session telemetry
Adaptive teaching loop trigger rate	Track per concept	Count of remediation loops

Parent satisfaction score	≥4.2/5.0	In-app parent feedback survey

3. Feature Overview & User Flow

3.1 High-Level Flow

The “Tutor Me” feature follows a structured pedagogical loop for each lesson:

1. **Trigger:** Student taps “Tutor Me” button visible during Lesson Shorts video playback.
2. **Session Initialisation:** Video pauses. Miss Lily AI Tutor session opens in a dedicated interactive view. System retrieves all segment data (images, text scripts, animations) from Smart Content Studio database for the current video.
3. **Segment-by-Segment Teaching Loop:** For each segment of the video (Segment 1 through Segment N):
    - TEACH: Miss Lily explains the segment using the retrieved images and text content, presenting them in an interactive conversational format.
    - ASSESS: Student is prompted to repeat/explain what they just learned in their own words (voice or text input).
    - EVALUATE: AI evaluates the student’s response against the segment’s key concepts.
    - REMEDIATE (if needed): If the student’s response is incomplete or incorrect, Miss Lily re-teaches using an adapted approach (simpler language, analogies, different visual emphasis) — still using the same content database assets. This loop repeats until the student demonstrates understanding.
    - ADVANCE: Once the student passes, proceed to the next segment.
4. **Final Comprehensive Assessment:** After all segments are completed, the student is asked to explain the entire lesson topic from start to finish.
5. **Final Evaluation & Report:** AI evaluates the full response, provides a mastery score, highlights strengths and weak areas, and logs results to the student’s mastery profile.

3.2 Visual Flow Diagram

The following represents the state machine for a single Tutor Me session:

```
[Video Playing] → Tap “Tutor Me” → [Session Init] → [Load Segment 1]

→ [Teach] → [Assess] → [Evaluate]
→ Pass? → Yes → [Next Segment] / No → [Remediate] → [Re-Assess]
→ [All Segments Done] → [Full Lesson Assessment] → [Final Score & Report]
```

4. Detailed Functional Requirements

4.1 FR-01: “Tutor Me” Button & Session Trigger

Attribute	Specification

Button Placement	Visible on the Lesson Shorts video player overlay, alongside existing controls (play/pause, progress bar, like, share). Positioned prominently — recommended: bottom-right area or as a floating action button.
Button Label	“Tutor Me” with the Miss Lily avatar icon
Availability	Available at any point during video playback. Also available after video completion on the end screen.
Trigger Behaviour	On tap: (1) Pause video at current frame. (2) Store the current playback timestamp. (3) Transition to the Tutor Me interactive session view with a smooth animation. (4) Begin loading segment data from Smart Content Studio.
Resume Capability	Student can exit the Tutor Me session at any time and resume video from the paused timestamp. Partial session progress is saved.

4.2 FR-02: Smart Content Studio Data Retrieval

Attribute	Specification
Data Source	Smart Content Studio database — the same source used during video generation.
Data Retrieved per Segment	Segment script text (narration transcript), background images / illustrations, animation assets, key concept tags/keywords, Bloom’s taxonomy level, curriculum alignment metadata (subject, grade, board, chapter, topic).
Retrieval Timing	Prefetch all segments for the current video when “Tutor Me” is triggered. Display a loading state (Miss Lily waving / preparing) during fetch.
Caching	Cache retrieved segment data locally for the duration of the session. If the student re-enters Tutor Me for the same video within the same app session, serve from cache.
Fallback	If any segment data is missing or corrupted, skip to the next available segment and log an error for admin review. Never show a broken state to the student.

4.3 FR-03: Segment Teaching Phase (“Teach”)

Attribute	Specification
Presenter	Miss Lily AI Tutor avatar — voice-interactive 3D avatar, consistent with the existing Miss Lily implementation.
Content Presentation	Miss Lily explains the segment’s concept conversationally. The original segment images and illustrations from the Smart Content Studio are

	displayed alongside her explanation. Text highlights or key terms appear on screen as she speaks.
Visual Layout	Split view: Miss Lily avatar on one side, segment images/illustrations/key text displayed on the other side. Images should animate in sequentially as Miss Lily references them.
Voice & Language	Miss Lily speaks in the student’s configured language (Tamil or English, Phase 2: French). Voice tone is encouraging and age-appropriate for 8–16 year olds.
Pacing	Student can tap to pause Miss Lily mid-explanation. A “Repeat” button allows the student to hear the segment explanation again before the assessment phase.
Segment Indicator	A progress bar or step indicator shows: “Segment 2 of 5” so students know how far they are in the lesson.

4.4 FR-04: Student Response Phase (“Assess”)

Attribute	Specification
Prompt	After the teaching phase, Miss Lily asks: “Now, can you explain what you just learned about [segment topic] in your own words?” The prompt is contextualised to the specific segment.
Input Methods	Voice input (primary, via device microphone — speech-to-text), typed text input (secondary, on-screen keyboard), handwritten answer via stylus/touch (scanned by AI grading, on supported devices).
Visual Aids During Response	The segment’s images remain visible on screen during the student’s response as reference material. The student should explain using these images, not from rote memory of text alone.
Time Allowance	No hard time limit. A gentle nudge (“Take your time!”) if no input is detected after 30 seconds. A second nudge after 60 seconds (“Would you like me to explain this again?”) with option to re-hear the segment.
COPPA Compliance	No raw voice data stored beyond the active session. Voice is converted to text in real-time, evaluated, and discarded. Only the evaluation result and mastery score are persisted.

4.5 FR-05: AI Evaluation Engine (“Evaluate”)

Attribute	Specification
Evaluation Criteria	The AI compares the student’s response against the segment’s key concept tags and expected understanding level (from Bloom’s taxonomy metadata). Evaluation checks for: (1) Coverage: Did the student mention all key concepts? (2) Accuracy: Are the facts correct? (3) Depth: Is the explanation at the expected Bloom’s level? (4) Coherence: Does the explanation make logical sense?
Scoring	Each segment is scored on a 0–100 scale. Pass threshold: ≥70/100. Scores are broken down by coverage, accuracy, depth, and coherence sub-scores.
Feedback to Student	If PASS: Miss Lily gives positive, specific encouragement (“Great job! You nailed how chlorophyll captures sunlight!”). If FAIL: Miss Lily identifies the specific gap without making the student feel bad (“You’re almost there! Let’s go over the part about chloroplasts again.”). Feedback is always encouraging and age-appropriate.
Gap Identification	The evaluation engine must identify exactly which key concepts the student missed or got wrong, so the remediation phase can target those specific gaps.

4.6 FR-06: Adaptive Remediation Phase (“Remediate”)

This is the core differentiator of the “Tutor Me” feature. When a student does not pass the segment evaluation, Miss Lily must adapt her teaching approach to help the student understand.

Attribute	Specification
Trigger	Automatically triggered when a student scores below 70/100 on a segment evaluation.
Targeted Re-teaching	Miss Lily does NOT repeat the entire segment. She focuses only on the specific concepts the student missed, as identified by the evaluation engine.
Adaptive Strategies	The AI selects from the following strategies based on the nature of the gap: (1) Simplification: Re-explain using simpler vocabulary and shorter sentences. (2) Analogy: Use a real-world analogy the student can relate to. (3) Visual Emphasis: Zoom into or highlight specific parts of the segment’s images. (4) Step-by-Step Breakdown: Break the concept into smaller sub-steps. (5) Question-Led: Ask leading questions to guide the student to the answer themselves.
Content Source	All remediation content must use the same images and text assets from the Smart Content Studio database. The AI may re-compose and reframe these assets but must NOT introduce content from outside the video’s source material.

Maximum Remediation Loops	Maximum 3 remediation attempts per segment. After 3 failed attempts, Miss Lily provides the full correct explanation, marks the concept as a “weak area” in the student’s profile, and advances to the next segment. The weak area is flagged for spaced repetition in future sessions.
Escalation Hints	On attempt 2, Miss Lily offers a partial hint. On attempt 3, she provides a more detailed scaffold. This graduated approach ensures the student is supported without being given the answer outright.

4.7 FR-07: Final Comprehensive Assessment

Attribute	Specification
Trigger	Automatically presented after the student has completed all segments of the lesson.
Prompt	Miss Lily says: “Amazing work going through each part! Now, can you explain the entire lesson about [topic] from beginning to end, in your own words?”
Input Methods	Same as segment assessment: voice, typed text, or handwritten input.
Visual Support	All segment images are displayed in a scrollable gallery/carousel so the student can reference them while explaining.
Evaluation Criteria	Holistic evaluation across ALL segments: completeness (did they cover all segments?), accuracy, logical flow (did they explain concepts in the right order?), depth per Bloom’s taxonomy level.
Scoring	Overall mastery score: 0–100. Breakdown by segment. Pass threshold: ≥60/100 overall.
If Student Fails	Miss Lily identifies the weakest segments and offers to re-teach those specific segments only (not the full lesson). After re-teaching, the student is given one more attempt at the final assessment.
Celebration on Pass	Animated celebration (confetti, Miss Lily clapping, XP/coin reward animation). Award XP and coins tied to the mastery score.

4.8 FR-08: Session Completion & Reporting

Attribute	Specification
Session Summary Screen	After the final assessment, display a summary card showing: overall mastery score, per-segment scores, time spent, number of remediation loops triggered, XP and coins earned, weak areas flagged for review.

Data Logging	Log to student’s mastery profile: per-concept mastery scores, concepts marked as weak, number of attempts per segment, Bloom’s taxonomy level achieved, session duration.
Teacher Dashboard	Tutor Me session results feed into the Teacher Analytics Dashboard: class-wide mastery heatmap updated in real-time, individual student progress, concept difficulty trends (which segments cause the most remediation loops).
Parent Dashboard	Parent sees: “Your child completed an AI tutoring session on Photosynthesis and scored 85/100.” Breakdown of time in learning mode vs. gaming mode.
Spaced Repetition	Weak concepts identified during the Tutor Me session are automatically queued into the Adaptive Question Engine for spaced repetition in future game sessions and assessments.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Tutor Me session must initialise within 3 seconds of button tap (including data fetch). AI evaluation of student response must complete within 2 seconds.
NFR-02	Offline Support	Phase 1: Online only. Phase 2: Allow pre-caching of segment data so Tutor Me can work in low-connectivity environments (critical for semi-urban and rural India).
NFR-03	Scalability	Must support concurrent Tutor Me sessions for up to 10,000 students simultaneously without degradation (targeting Year 1: 5,000 paying students).
NFR-04	Platform Support	Phase 1: Mobile (iOS & Android). Phase 2: Desktop (Windows & Mac), Gaming consoles, VR headsets. UI must be responsive and adapt to each form factor.
NFR-05	Child Safety	COPPA compliant: no raw voice data persisted. India DPDP Act 2023 compliant: no child PII shared. All AI responses are guardrailed against inappropriate content.
NFR-06	Accessibility	Subtitles for all Miss Lily speech. High-contrast mode support. Font size adjustable. Screen reader compatibility for typed interactions.
NFR-07	Language	Phase 1: Tamil and English. Phase 2: French. All AI tutoring, prompts, evaluation feedback, and UI labels must be fully localised.
NFR-08	Bandwidth	Optimised for low-bandwidth connections (target: functional at 256 kbps). Segment images should be pre-compressed. Voice processing should use edge inference where possible.

6. Data Model & API Requirements

6.1 Smart Content Studio — Data Schema per Video Segment

The following data must be retrievable from the Smart Content Studio database for each segment of a Lesson Short video:

Field	Type	Description
video_id	String (UUID)	Unique identifier for the Lesson Short video

segment_index	Integer	Ordinal position of the segment within the video (1, 2, 3...)
segment_title	String	Short title of the segment (e.g., “What is Photosynthesis?”)
narration_script	Text (Long)	Full narration text used by the avatar in this segment
key_concepts	Array[String]	Tagged key concepts (e.g., [“chlorophyll”, “sunlight”, “carbon dioxide”])
images[]	Array[Object]	Each object contains: image_url, alt_text, display_order, caption
blooms_level	Enum	Remember | Understand | Apply | Analyse | Evaluate | Create
curriculum_metadata	Object	{ board, grade, subject, chapter, topic }
expected_explanation	Text	Model answer the AI uses as the evaluation rubric for this segment
timestamp_start	Float (seconds)	Start time of this segment in the original video
timestamp_end	Float (seconds)	End time of this segment in the original video

6.2 Tutor Me Session Data Model

A new data model is required to store each Tutor Me session:

Field	Type	Description
session_id	String (UUID)	Unique session identifier
student_id	String (UUID)	Reference to student profile
video_id	String (UUID)	Reference to the Lesson Short video
started_at	DateTime	Session start timestamp
completed_at	DateTime (nullable)	Session completion timestamp (null if abandoned)
segment_results[]	Array[Object]	Per-segment: { segment_index, score, attempts, passed, weak_concepts[], time_spent }
final_assessment_score	Integer (0–100)	Score from the final comprehensive assessment
overall_mastery_score	Integer (0–100)	Weighted average of segment + final scores
xp_earned	Integer	XP awarded for this session

coins_earned	Integer	Coins awarded for this session
language	Enum	Language used in this session (Tamil / English / French)

6.3 Required API Endpoints

Method	Endpoint	Description
GET	/api/v1/videos/{video_id}/segments	Retrieve all segment data for a video from Smart Content Studio
POST	/api/v1/tutor-sessions	Create a new Tutor Me session
POST	/api/v1/tutor-sessions/{id}/evaluate	Submit student response for AI evaluation (per segment)
POST	/api/v1/tutor-sessions/{id}/final-evaluate	Submit final comprehensive response for evaluation
PATCH	/api/v1/tutor-sessions/{id}	Update session (mark complete, save progress)
GET	/api/v1/tutor-sessions/{id}/summary	Retrieve session summary and scores
POST	/api/v1/tutor-sessions/{id}/remediate	Request adaptive remediation content for failed concepts

7. UI/UX Requirements

7.1 Tutor Me Session Screen Layout

- **Miss Lily Avatar Area (30–40% of screen):** Displays the animated Miss Lily 3D avatar. Lip-syncs to speech. Expressive reactions (nodding, smiling, thinking pose) based on context.
- **Content Display Area (50–60% of screen):** Shows segment images, illustrations, and highlighted key text from Smart Content Studio. Images animate in/out as Miss Lily references them. Supports pinch-to-zoom on images.
- **Progress Indicator:** Persistent progress bar showing current segment out of total. Completed segments marked with green checkmarks. Failed segments marked with amber retry icon.
- **Input Area:** Expandable from bottom of screen. Microphone button (primary) with recording animation. Text input field (secondary). Camera/stylus input button (for handwritten answers).
- **Control Bar:** Exit button (with "Save & Exit?" confirmation). Repeat button (re-hear current segment explanation). Pause button. Help button (get a hint without triggering full remediation).

7.2 Interaction Design Principles

- All interactions must feel like a conversation, not a test. Miss Lily's tone is that of an encouraging older sister, not a strict examiner.
- Transitions between phases (Teach → Assess → Evaluate → Remediate) should be smooth and conversational, not abrupt state changes.
- Celebrate every small win. Even partial understanding should receive positive acknowledgement before correction.
- The UI must be thumb-friendly for mobile — all primary actions reachable with one hand on a smartphone.
- Dark mode and light mode support, consistent with the existing PTL AI app theme.

8. Gamification & Reward Integration

The Tutor Me feature must integrate with PTL AI's existing reward system:

Action	XP Reward	Coin Reward
Complete a segment on first attempt	+50 XP	+10 Coins
Complete a segment after remediation	+30 XP	+5 Coins
Pass final comprehensive assessment	+200 XP	+50 Coins
Score 90+ on final assessment	+100 Bonus XP	+25 Bonus Coins

Complete full Tutor Me session	+100 XP	+20 Coins
Streak: 3 consecutive sessions in a week	+500 Bonus XP	+100 Bonus Coins

9. Edge Cases & Error Handling

Scenario	Handling
Student exits mid-session	Save progress. On re-entry for the same video, offer: “Resume where you left off?” or “Start over?”
Network drops during session	Cache current segment data. Allow offline evaluation for the current segment using on-device model. Sync results when connectivity returns.
Student gives irrelevant response	Miss Lily gently redirects: “Hmm, that’s interesting, but let’s focus on [topic]. Can you tell me about [specific concept]?” Count as a failed attempt.
Student gives empty/silent response	After 30s: gentle nudge. After 60s: offer to re-explain. After 90s: offer to skip to next segment with the concept marked as weak.
Video has no segment data in database	Disable the “Tutor Me” button for this video. Show tooltip: “Tutor Me coming soon for this lesson.”
Student is in a noisy environment	If voice recognition confidence is below threshold, prompt: “I’m having trouble hearing you. Would you like to type your answer instead?”
Student repeatedly fails all segments	After 3 complete remediation failures across 2+ segments, Miss Lily suggests: “Why don’t you watch the video once more and come back to practice?” Redirect to video replay.
Multiple language switching	Student’s language preference is locked for the duration of a session. Language can only be changed before starting or when starting a new session.

10. Integration Points with Existing PTL AI Systems

System	Integration
Smart Content Studio	Source of all teaching content (images, text, scripts). The Tutor Me feature reads from the same database tables the video generation engine writes to. No separate content authoring required.

Miss Lily AI Tutor	Existing 3D avatar and voice engine. Tutor Me extends Miss Lily's capabilities from free-form Q&A to structured segment-based teaching. Reuses the same avatar rendering pipeline, voice synthesis, and emotion detection.
Adaptive Question Engine	Weak concepts flagged by Tutor Me are fed into the Adaptive Question Engine for spaced repetition. The engine's Bloom's taxonomy tagging is shared with Tutor Me's evaluation criteria.
Gamification System	XP, coins, and streak tracking. Tutor Me sessions contribute to daily learning time (not gaming time) in parent analytics.
Teacher Analytics Dashboard	Tutor Me session results populate: class-wide concept mastery heatmap, per-student progress timeline, concept difficulty analytics (which segments trigger the most remediation).
Parent Dashboard	Session completion notifications. Mastery score display. Learning time attribution.
Student Mastery Profile	Per-concept mastery scores updated after each session. Weak areas flagged for future review. Historical session data for learning trajectory visualisation.

11. Phased Rollout Plan

Phase 1 — MVP (Month 1–3)

- Tutor Me button on Lesson Shorts video player (mobile only: iOS & Android)
- Segment-by-segment teaching with images and text from Smart Content Studio
- Voice and text input for student responses
- AI evaluation with pass/fail per segment
- Basic remediation: re-explain with simplified language
- Final comprehensive assessment
- Session summary with mastery score
- XP and coin rewards
- Tamil and English language support

Phase 2 — Enhanced Adaptivity (Month 4–6)

- Advanced adaptive remediation strategies (analogy, visual emphasis, question-led)
- Handwritten answer scanning and evaluation
- Offline/low-connectivity support with edge inference
- Desktop platform support (Windows & Mac)
- Teacher Dashboard integration with Tutor Me analytics

- Spaced repetition queue integration with Adaptive Question Engine

Phase 3 — Scale & Intelligence (Month 7–12)

- Gaming console and VR headset support
- French language support
- Emotion detection: Miss Lily adjusts tone based on student's facial expressions/voice tone
- Predictive difficulty: AI pre-adjusts teaching approach based on the student's historical mastery profile
- Peer learning: option to pair students for collaborative Tutor Me sessions (school-verified buddies only)
- A/B testing framework for remediation strategy effectiveness

12. Dependencies & Risks

ID	Risk / Dependency	Impact	Mitigation
D-01	Smart Content Studio must have segment-level data for all videos	Feature cannot work without segment data	Audit existing video library. Flag videos without segments. Prioritise backfilling.
D-02	AI evaluation accuracy for children's speech patterns	Poor evaluation leads to frustration	Train evaluation model on child speech corpus. Generous scoring in Phase 1. Iteratively tighten.
R-01	AI API costs per session at scale	Unit economics impacted if AI calls are expensive	Use edge inference for speech-to-text. Batch evaluation calls. Monitor cost per session.
R-02	Student fatigue in long lessons with many segments	Abandonment mid-session	Cap at 7 segments per session. Offer "save & resume". Gamification rewards for milestones.
R-03	Low-bandwidth regions may struggle with real-time AI evaluation	Unusable in rural/semi-urban areas	Phase 2 offline mode. Compress assets. Edge inference for basic evaluation.
R-04	Content guardrails for AI-generated remediation responses	AI could generate inappropriate content for children	All AI responses pass through content safety filter. Only use source material from Smart

			Content Studio. No open-ended generation.

13. Acceptance Criteria

The following criteria must be met for each phase to be considered complete:

Phase 1 MVP — Acceptance Criteria

1. Student can tap “Tutor Me” during any Lesson Short video that has segment data in Smart Content Studio.
2. Video pauses and Miss Lily AI Tutor session opens within 3 seconds.
3. All segment images, text, and scripts are correctly retrieved and displayed from the Smart Content Studio database.
4. Miss Lily explains each segment using the retrieved content with voice narration and on-screen visuals.
5. Student can respond via voice or text after each segment.
6. AI correctly evaluates student responses against key concepts with ≥80% alignment to human evaluator scores.
7. Students who fail a segment receive a simplified re-explanation and can re-attempt.
8. After all segments, a final comprehensive assessment is presented and evaluated.
9. Session results are saved to student’s mastery profile and visible in parent dashboard.
10. XP and coins are correctly awarded per the reward table.
11. No raw voice data is persisted beyond the active session (COPPA compliance).
12. Feature works on iOS and Android devices across screen sizes (phones and tablets).

End of Document

For questions or feedback on this PRD, contact Thiruvengadam at thiru@imercfy.com.